Nonparametric maximum likelihood estimation of intensities for covariate-driven Poisson processes

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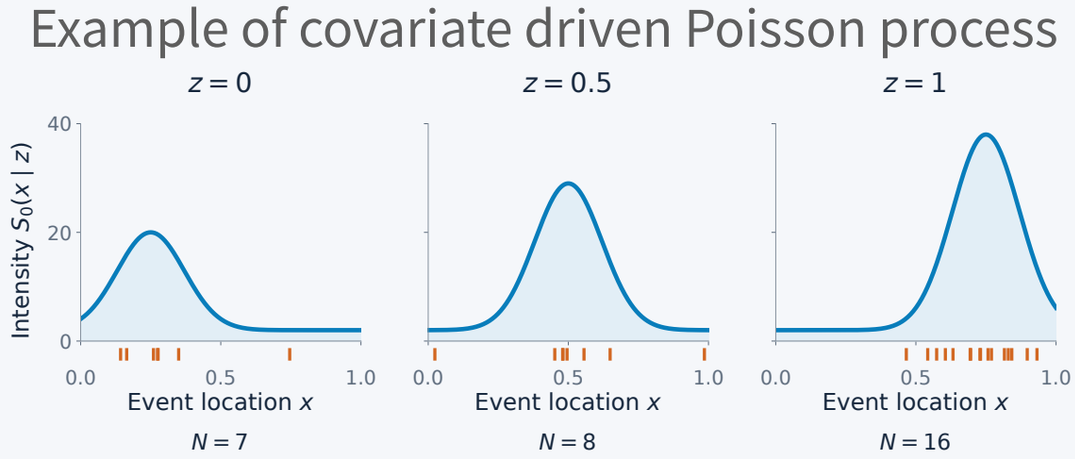
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Overview

★Covariate-driven Poisson processes

$$Z \sim P_Z,\ X \mid Z \sim \text{Poisson process}(S(\cdot \mid Z))$$

- Covariates change the event intensity
- Estimating the intensity function $S(\cdot \mid \cdot)$ is crucial in applications
- We consider theoretical properties of the MLE of the intensity function



- Observations: $\{(Z_i, X_i)\}_{i=1}^n$

★Problem: Vanishing fitted intensities can make KL risk infinite.

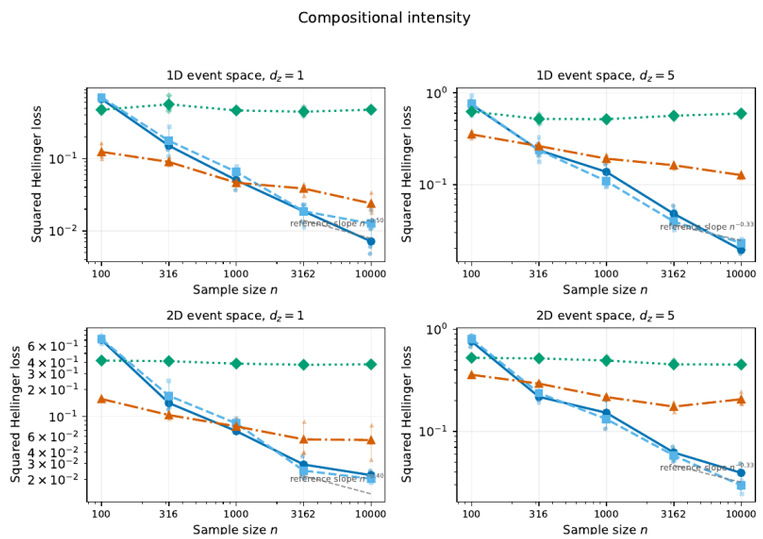
★Approach: Fit by maximum likelihood; evaluate squared Hellinger loss.

★Oracle inequality: Separate estimation complexity from approximation error.

★Convergence rates: Exploit compositional structure or intrinsic dimension.

★Optimality: Minimax up to log factors for the Euclidean compositional class (α = 0).

★Numerical experiments



Poisson processes

Poisson Processes and Intensity Functions

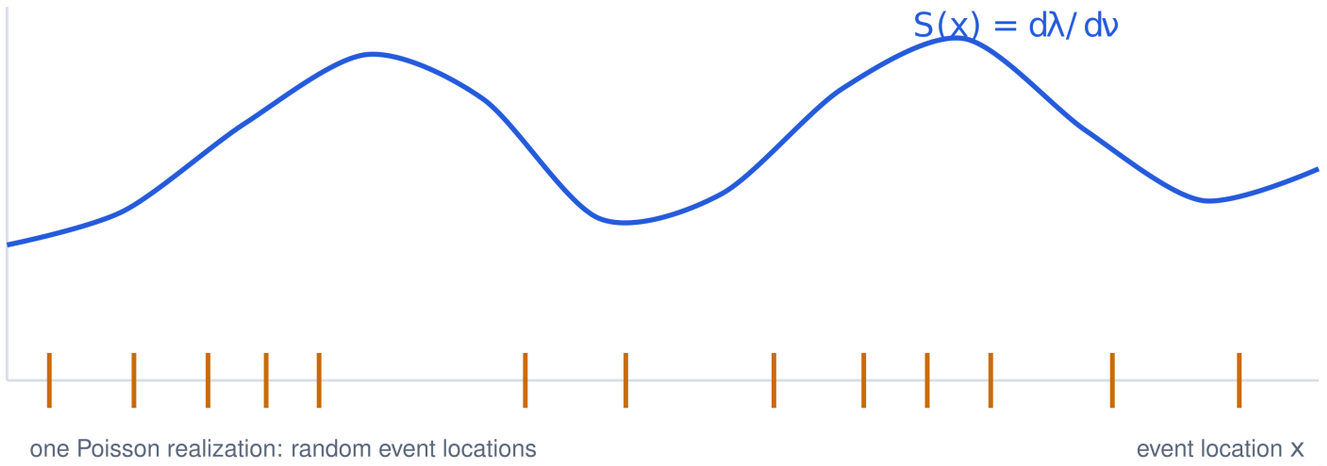
- Poisson processes model random events occurring in time or space
- Key features of a Poisson process X
  - X is a nonnegative integer-valued random measure.
    X(B) counts the events in a measurable set B
  - Counts on disjoint measurable sets are mutually independent:

$$B_i \cap B_j = \varnothing \quad \Rightarrow \quad X(B_i) \perp\!\!\!\perp X(B_j)$$

- The number of events on a measurable region follows the Poisson law: $X(B) \sim \mathrm{Poi}\{\int_B S(x)\, d\nu(x)\}$

Intensity function

- The intensity function controls expected counts and the distribution of events



Poisson Processes with Covariates

⭐Conditional intensity functions

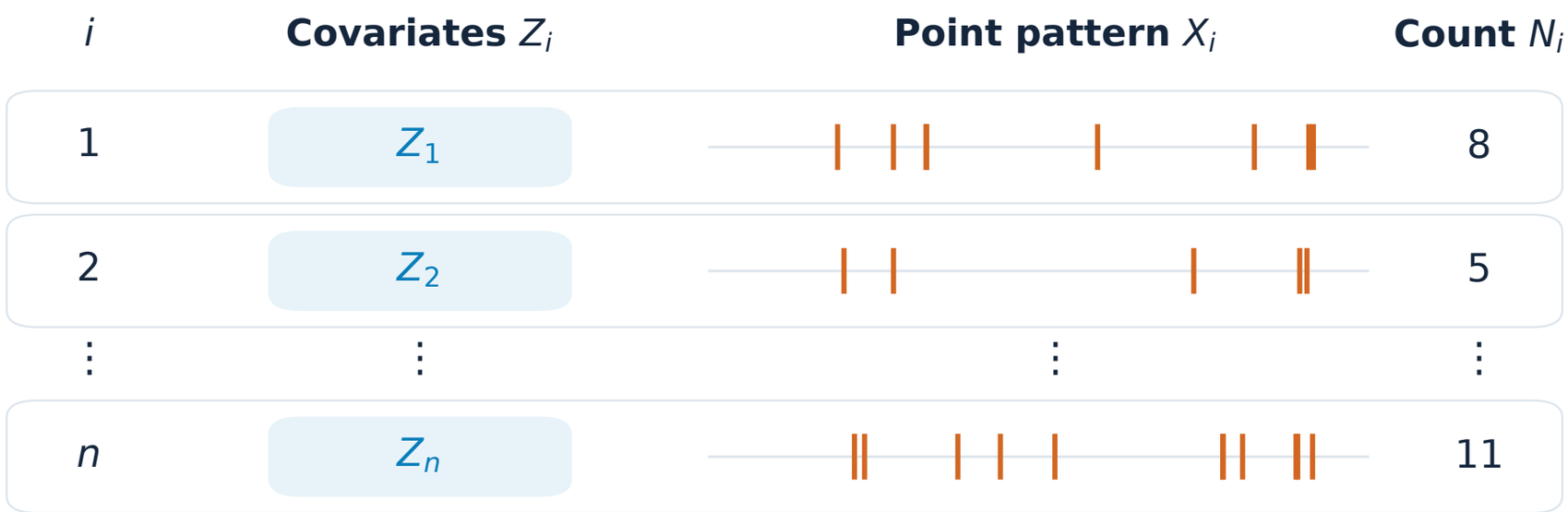
- We consider intensity functions that depend on the covariates Z

$$X \mid Z \sim \text{Poisson process}(S(\cdot \mid Z))$$

Conditional intensity function

⭐Why condition on Z?

- In real data, intensities naturally change due to many factors
- The intensity function may change its height, location, or shape with covariates Z



$$X_i \mid Z_i = z \;\sim\; \mathrm{PP}\{S_0(x \mid z)\, d\nu(x)\}$$

n independent pairs

Maximum Likelihood Estimation for Conditional Intensities

⭐Nonparametric maximum likelihood estimator (NPMLE)

- $A \subset \mathbb{R}^{d_x}$: space of events, $[0,1]^{d_z}$: space of covariates, $d = d_x + d_z$
- $(Z_1, X_1), \ldots, (Z_n, X_n)$: i.i.d. covariate–point-pattern pairs

$$Z_i \sim_{\text{i.i.d.}} P_Z, \quad X_i \mid Z_i \sim \text{Poisson process}(S_0(\cdot \mid Z_i))$$

- S_0: underlying (true) intensity function
- Negative log-likelihood:

$$\ell_n(X_1, \ldots, X_n, \boldsymbol{z}_1, \ldots, \boldsymbol{z}_n \mid S) \coloneqq \frac{1}{n} \sum_{i=1}^{n} \left\{ -\sum_{j=1}^{N_i} \log(S(\boldsymbol{x}_{ij} \mid \boldsymbol{z}_i)) + \int_A S(\boldsymbol{x} \mid \boldsymbol{z}_i)\, d\nu(\boldsymbol{x}) \right\}$$

$\boldsymbol{x}_{ij}$: location of event j in X_i

$N_i = X_i(A)$, ν: measure on A

- $\mathcal{F}_n$: a class of nonnegative intensity functions
  - Here, we use ReLU neural networks.
- Nonparametric maximum likelihood estimator (NPMLE):

$$\hat{S}_n \in \underset{S \in \mathcal{F}_n}{\arg\min}\, \ell_n(S)$$

- The integral can be approximated via Monte Carlo

Difficulties in Nonparametric Maximum Likelihood Estimation

★Excess risk analysis of MLE in various settings

- **Regression**
  - Under Gaussian errors with fixed variance, excess likelihood risk is proportional to integrated squared error.
  - If both the true and fitted regression functions are bounded, the integrated squared error is finite.
  - **This boundedness assumption is reasonable**
- **Classification**
  - The excess risk involves the **KL divergence** between the true and estimated conditional probabilities
    - **KL risk can be infinite.**
  - **To evaluate the excess risk, we need additional assumptions**
    - e.g., the true conditional probability is bounded away from zero

$$c < \mathbb{P}(Y = 1 \mid \boldsymbol{x}) < C, \quad 0 < c, C < 1$$

- **Estimating intensity functions**
  - **As in classification, KL risk can be infinite.**

Difficulties in Nonparametric Maximum Likelihood Estimation

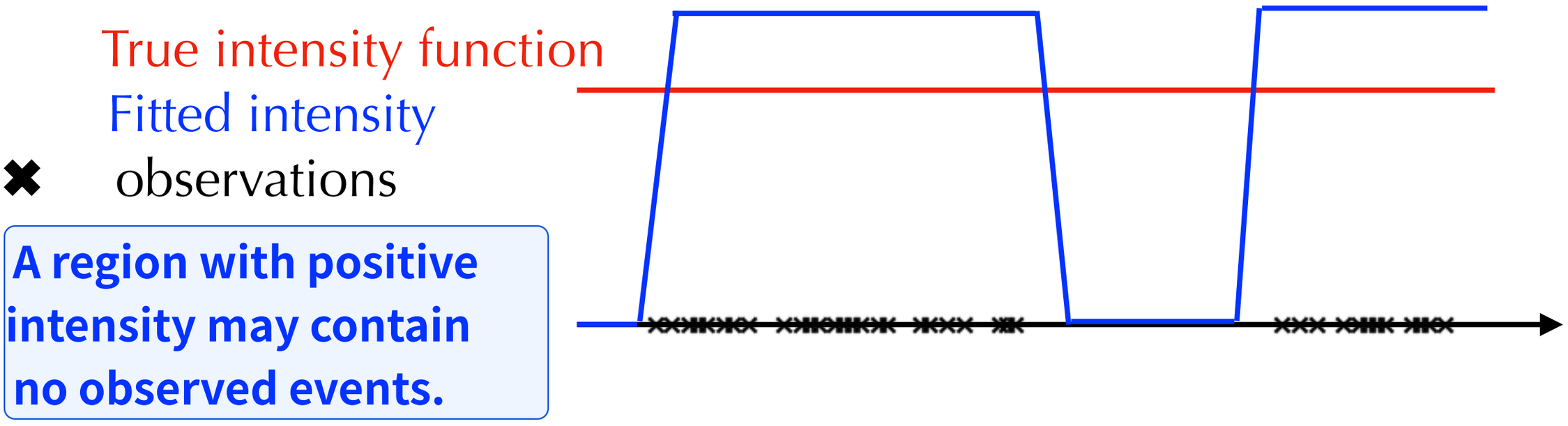
★Example where the KL divergence diverges

- For Poisson processes, the KL divergence involves **the ratio of intensity functions**:

$$\mathrm{KL}(P_f, P_g) = \int_A \left(f \log\left(\frac{f}{g}\right) + g - f \right) d\nu$$

P_h : the distribution of a Poisson process with intensity function h

- If the fitted intensity is zero on a set with positive true intensity measure, the KL divergence is infinite.



A region with positive intensity may contain no observed events.

Goals of This Study

- **To clarify the theoretical properties of the NPMLE of intensity functions for Poisson processes with covariates**
  - Issue: KL risk can be infinite.
  - Approach: Fit by maximum likelihood and evaluate squared Hellinger loss.

- **Analyze deep-learning–based NPMLE**
  - Exploits compositional structure and achieves minimax rates up to logarithmic factors for $\alpha = 0$ on the Euclidean cube.
  - On a known smooth manifold, the rate exponent uses the sum of the intrinsic and covariate dimensions.

Notation and Assumptions

⭐Notations

- $\tilde{H}^2(S_1, S_2) := \mathbb{E}_Z \left[\frac{1}{2} \int_A (\sqrt{S_1(\boldsymbol{x} \mid Z)} - \sqrt{S_2(\boldsymbol{x} \mid Z)})^2 d\nu(\boldsymbol{x}) \right]$
  - **Instead of KL divergence between Poisson processes, we evaluate $\tilde{H}^2$**
- $\mathcal{F}(L, r, B, F)$: class of ReLU networks
  - L : depth, r : width per layer, B : parameter magnitude bound
  - F : bound of the output
  - Activations: ReLU everywhere, including the output layer

⭐Assumption

$$\exists c_0 > 0,\ \forall n \in \mathbb{N},\ \exists S_n^* \in \mathcal{F}_n,$$

$$\forall \boldsymbol{x} \in A,\ \forall \boldsymbol{z} \in [0,1]^{d_z} :\quad S_0(\boldsymbol{x} \mid \boldsymbol{z}) \leq c_0^2 S_n^*(\boldsymbol{x} \mid \boldsymbol{z}).$$

- This holds automatically when using the DNN estimator

Oracle Inequality

★Oracle inequality

Theorem1 (Oracle Inequality)

For a uniformly bounded class satisfying the domination and local bracketing conditions, choose δ_n such that:

$$\sqrt{n}\delta_n^2 \geq c\Psi(\delta_n)$$

Then the following bound holds, with C independent of n.

Estimation error

$$\mathbb{E}[\tilde{H}^2(\hat{S}_n, S_0)] \leq C\left\{\delta_n^2 + \frac{1}{n} + \underline{\tilde{H}^2(S_n^*, S_0)}\right\}$$

Approximation error

- Ψ: model complexity
- The NPMLE convergence rate depends on
  the estimation error term + the approximation error term

Oracle Inequality

The upper bound depends on complexity and approximation error

- The upper bound, ignoring constants:

$$\mathbb{E}[\tilde{H}^2(\hat{S}_n, S_0)] \lesssim \delta^2 + \frac{c^3}{n} + \tilde{H}^2(S_n^*, S_0)$$

1 Complexity

$$\sqrt{n}\,\delta_n^2 \gtrsim \Psi(\delta_n).$$

Ψ bounds the local bracketing entropy integral.

2 Approximation

$$S_n^\star \in \mathcal{F}_n, \qquad \frac{S_0}{S_n^\star} \le c_0^2.$$

S_n^* may depend on the unknown truth; choose it as close to S_0 as possible.

Complexity ↑ | **Richer model class** | **Approximation ↓**

Choose the network architecture to balance estimation and approximation errors.

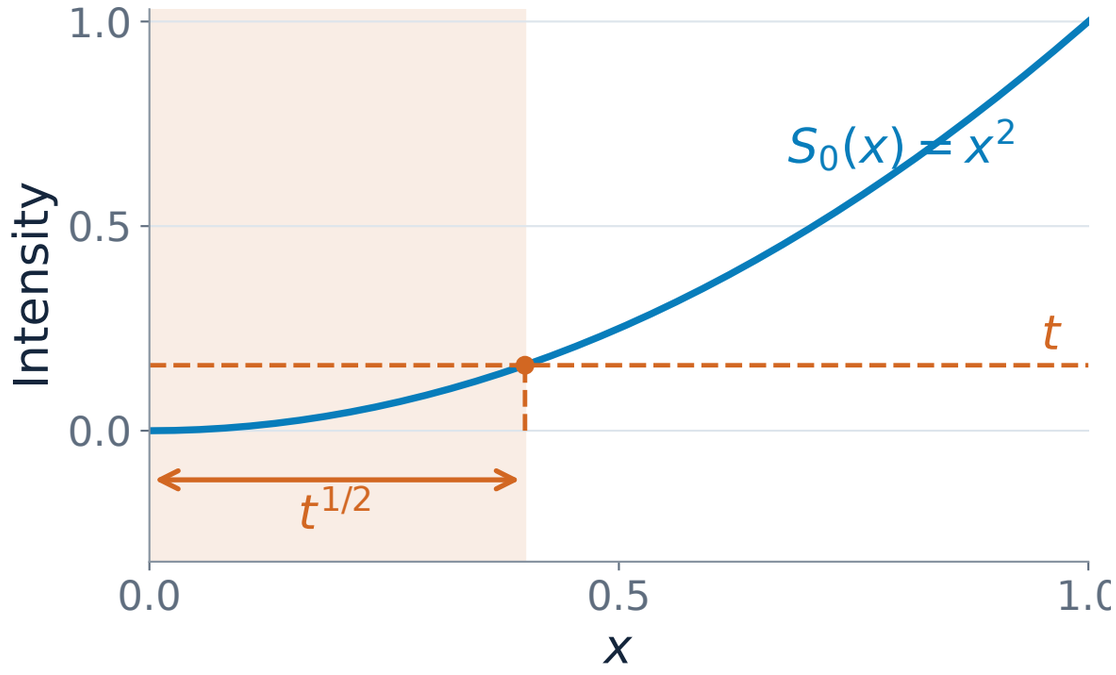
Small Value Bound (SVB)

⭐Small Value Bound condition improves approximation error

Uniform over the function class

$$V_S(t) := \int_A \mathbb{P}_Z(S(x \mid Z) \le t)\, d\nu(x) \le Ct^{\alpha}, \qquad t > 0.$$

Larger α ⇒ the near-zero region shrinks faster



From uniform approximation to Hellinger error

Under the α-SVB condition, for sufficiently large M:

$$\|S - S_0\|_\infty \le C_1 M^{-1}, \qquad S \ge M^{-1}$$

$$\Longrightarrow \quad \widetilde{H}^2(S, S_0) \lesssim M^{-\{1+(\alpha\wedge 1)\}}(1 + \log M).$$

α = 0: no restriction | α > 0: no positive-volume zero set | rate gain saturates at α = 1

Structure of the True Intensity Function

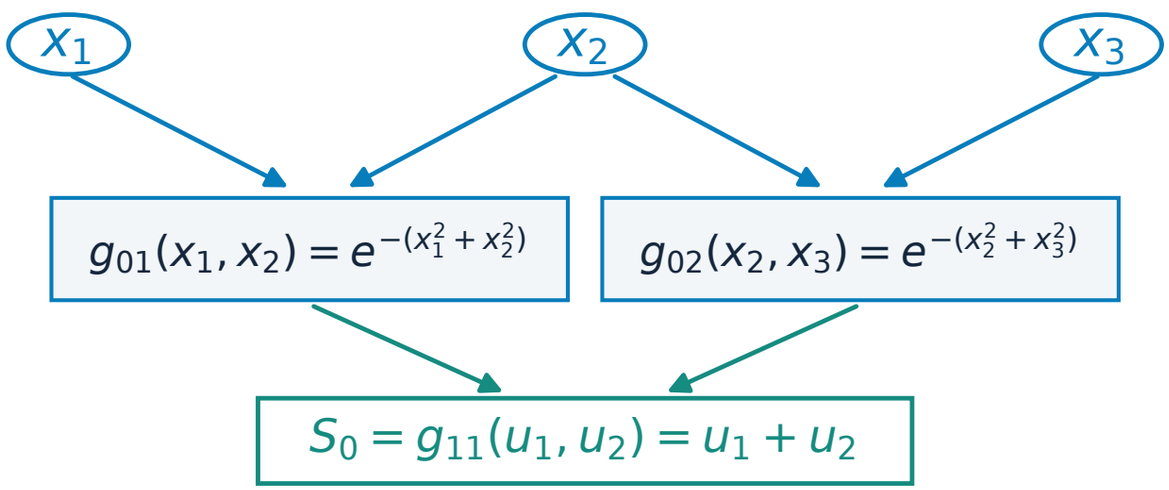
Compositional structure

- Hölder class with smoothness β, domain D, and bound Q: $\mathcal{C}^{\beta}(D, Q)$
- Compositional function class:

$$\mathcal{G}_{\text{comp}}(q, \mathbf{d}, \mathbf{t}, \boldsymbol{\beta}, Q) = \{f = \boldsymbol{g}_q \circ \cdots \circ \boldsymbol{g}_0 : \boldsymbol{g}_i = (g_{ij})_j : [a_i, b_i]^{d_i} \to [a_{i+1}, b_{i+1}]^{d_{i+1}},$$
$$g_{ij} \in \mathcal{C}^{\beta_i}([a_i, b_i]^{t_i}, Q), \text{for some} |a_i|, |b_i| \leq Q\}$$
$$\beta_i^* := \beta_i(\beta_{i+1} \wedge 1) \cdots (\beta_q \wedge 1).$$

- $f(x_1, x_2, x_3) = g_1 \circ \boldsymbol{g}_0(x_1, x_2, x_3) = g_{11}(g_{01}(x_1, x_2), g_{02}(x_2, x_3))$

3 dim. 2 dim.



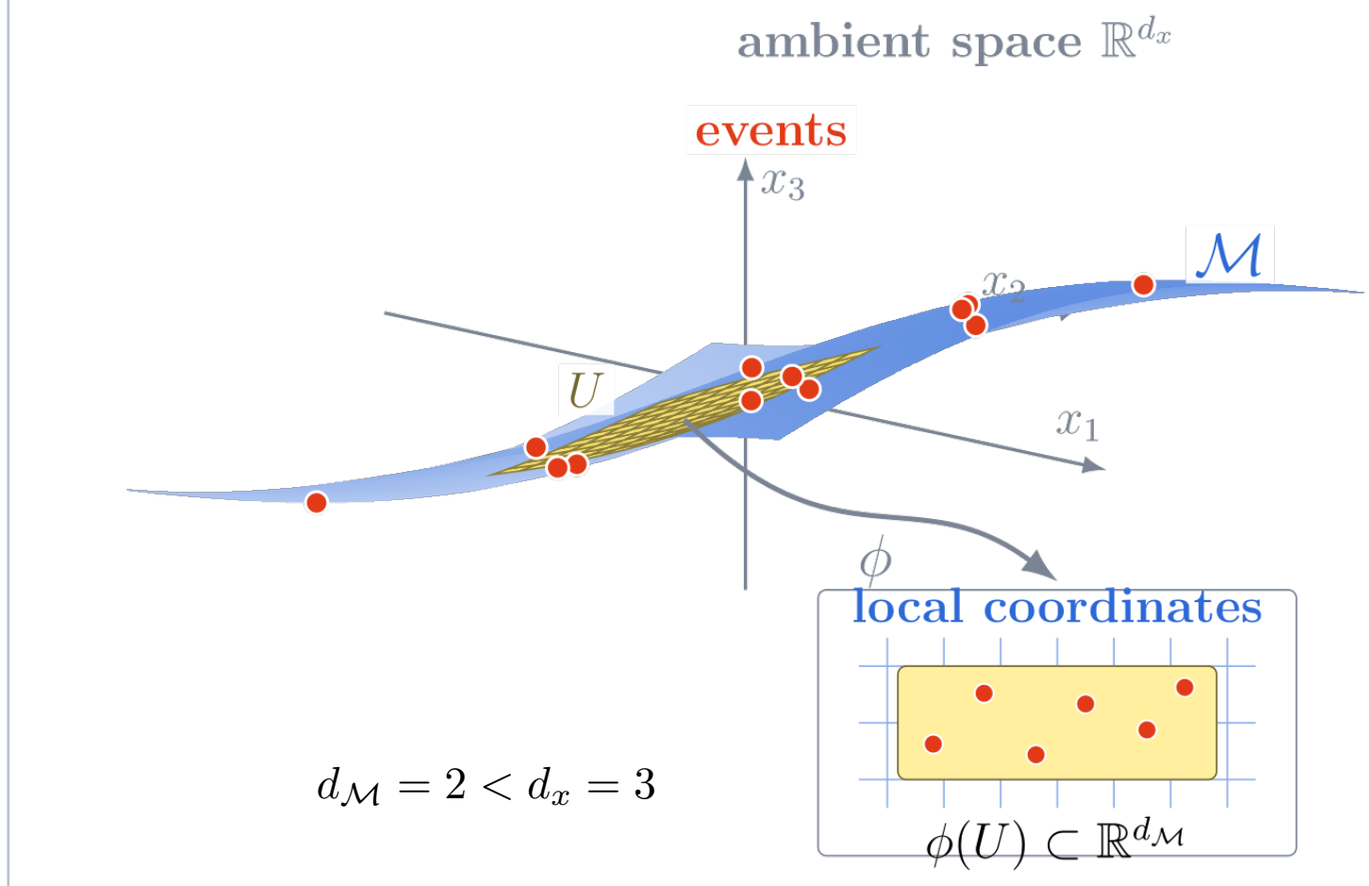
$d = 3, \quad t_0 = t_1 = 2$

- **Each component has two inputs; the full function has three.**

Structure of the True Intensity Function

⭐Manifold-supported intensities

- The manifold and its Riemannian volume measure are known.
  - $\mathcal{C}^{\beta}(\mathcal{M} \times [0,1]^{d_z})$
  - $\mathcal{M} \subset A$ is a compact Riemannian manifold with $d_{\mathcal{M}} < d_x$
    - $\dim(\mathcal{M}) = d_{\mathcal{M}}$



Convergence Rates for Deep Learning Estimators

⭐Convergence rates for DNN-based MLE

- Using Theorem 1, we obtain convergence rates for fixed function classes of the true intensity function

function class	$\mathcal{C}^{\beta}(\mathcal{M} \times [0,1]^{d_z}, Q)$	$\mathcal{G}_{\text{comp}}$
Exponent of n (log factors omitted)	$-\dfrac{\beta}{\beta + d_{\mathcal{M}} + d_z}$	$-\min_{i=0,\ldots,q} \dfrac{\beta_i^*}{\beta_i^* + t_i}$

When $\alpha = 0$

- **Rate exponents use component or intrinsic dimensions, rather than the ambient input dimension d**

⭐Why is the exponent different from the classical rate?

If S_0 is bounded away from zero

$$S_0 \geq c > 0: \quad (\sqrt{S} - \sqrt{S_0})^2 \lesssim |S - S_0|^2$$

If S_0 may equal zero

$$S_0 = 0: \quad (\sqrt{S} - \sqrt{S_0})^2 = |S - S_0|$$

At zeros, Hellinger approximation error is not quadratic.

$$\alpha = 0: \; n^{-\frac{\beta}{\beta + d_{\text{eff}}}} \quad \Longleftrightarrow \quad \alpha \geq 1: \; n^{-\frac{2\beta}{2\beta + d_{\text{eff}}}}$$

Minimax Optimality

⭐Minimax lower bound

Benchmark rate

$$r_n := \max_{i=0,\dots,q} n^{-\beta_i^*/(\beta_i^*+t_i)},$$
$$\beta_i^* := \beta_i(\beta_{i+1} \wedge 1) \cdots (\beta_q \wedge 1).$$

Each t_i is the active input dimension of g_i

Minimax risk

$$R_n(\mathcal{G}) := \inf_{\widetilde{S}_n} \sup_{S_0 \in \mathcal{G}} \mathbb{E}_{S_0}\left[\widetilde{H}^2(\widetilde{S}_n, S_0)\right].$$

Smallest worst-case risk over all estimators

Bounds match for the Euclidean compositional class (α = 0).

$$c\, r_n \;\le\; R_n(\mathcal{G}_{\text{comp}}) \;\le\; \sup_{S_0 \in \mathcal{G}_{\text{comp}}} \mathbb{E}_{S_0}\left[\widetilde{H}^2(\widehat{S}_n, S_0)\right] \;\le\; C\, r_n \log^3 n.$$

No estimator is uniformly faster

DNN-NPMLE attains the same exponent

- **The rate is minimax optimal up to logarithmic factors.**

Numerical Experiments

⭐Design

$$Z_i \sim \mathrm{Unif}([0,1]^{d_z}), \qquad X_i \mid Z_i = z_i \sim \mathrm{PP}(S_0(\cdot \mid z_i)), \quad i = 1, \ldots, n.$$

PP = Poisson point process

Independent replicates	$n = 100,\ 316,\ 1000,\ 3162,\ 10000$
Covariate dimension	$d_z = 0,\ 1,\ 5$
Event supports	Unit interval, unit square, circle, sphere
Intensity designs	Compositional, near-zero, and manifold settings
Mean event count	Approximately 30 per replicate
Monte Carlo repetitions	20 per setting
Estimators	DNN–Softplus, DNN–ReLU, and kernel baselines

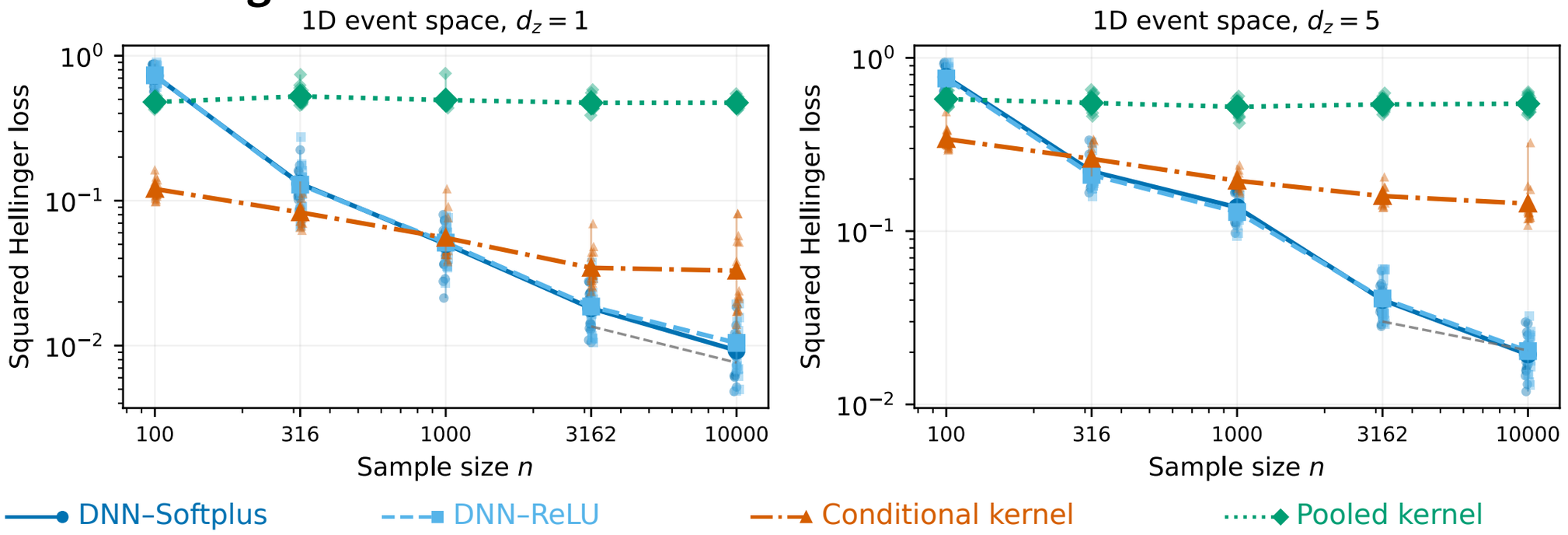
Loss $$\widetilde{H}^2(\widehat{S}, S_0) = \tfrac{1}{2}\,\mathbb{E}_Z\Big[\int\Big(\sqrt{\widehat{S}(x \mid Z)} - \sqrt{S_0(x \mid Z)}\Big)^2 d\nu(x)\Big]$$

Expected event count: approximately 30 per replicate.
The conditional kernel uses Z; the pooled kernel ignores Z.
Both DNNs use ReLU hidden layers, with either a Softplus or ReLU output.

Numerical Experiments

⭐ Results for compositional intensities

At n = 10,000, both DNNs have lower mean loss than the conditional kernel in all four settings.

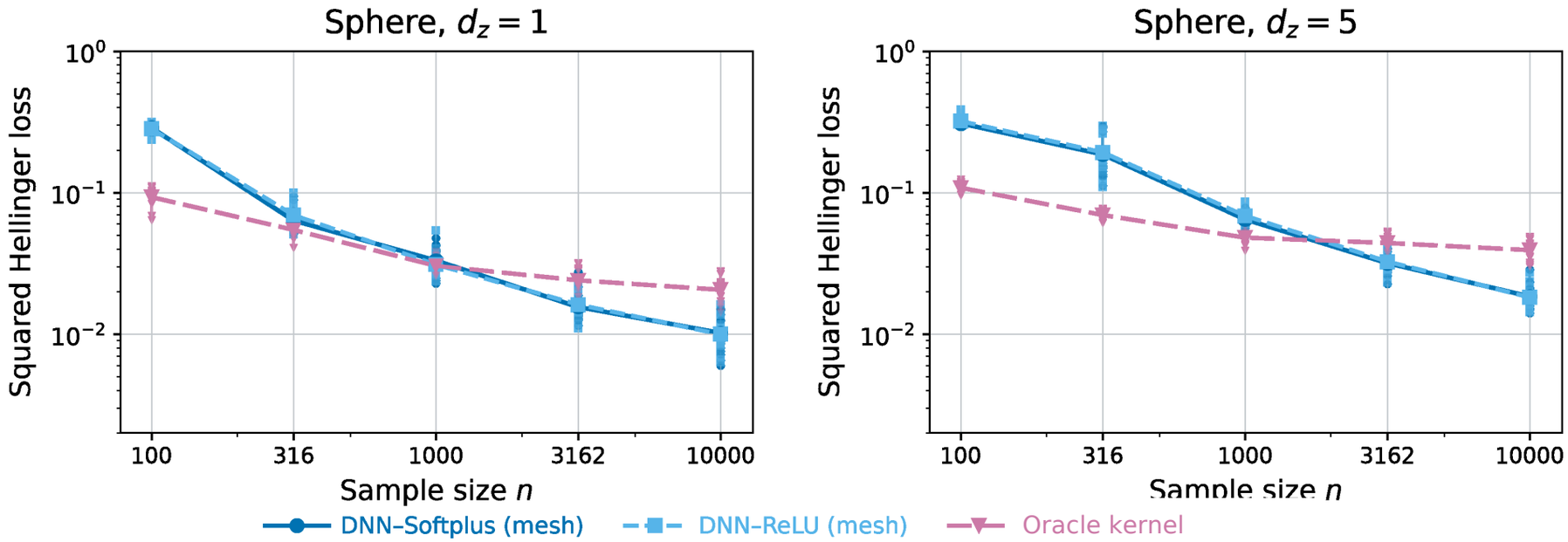


Design: event dimension, d_z	**DNN–Softplus**	**DNN–ReLU**	**Conditional kernel**
Composition: 1, 1	**0.00926**	0.0104	0.0329
Composition: 1, 5	**0.0194**	0.0203	0.144
Composition: 2, 1	**0.0205**	0.0213	0.0441
Composition: 2, 5	0.0346	**0.0329**	0.182

Numerical Experiments

⭐ Results for manifold-supported intensities

At n = 10,000, both DNNs have lower mean loss than the oracle kernel in three of the four settings.



Learning curves: sphere (exploratory).
Table: circle and sphere, n = 10,000.
DNNs: reconstructed integration meshes.
Oracle kernel: known geodesic geometry.
Lines: means; bars: ranges over 20 runs.

Mean squared Hellinger loss; $n = 10{,}000$; 20 repetitions

Support: intrinsic dim., d_z	**DNN–Softplus**	**DNN–ReLU**	**Oracle kernel**
Circle: 1, 1	0.00570	0.00643	**0.00520**
Circle: 1, 5	**0.0105**	0.0108	0.0244
Sphere: 2, 1	0.0102	**0.00997**	0.0205
Sphere: 2, 5	0.0184	**0.0182**	0.0392

Bold: smallest mean in each row.

Summary

- **We establish theoretical guarantees for NPMLE of covariate-driven Poisson intensities (Theorem 1)**

- **Convergence rates and numerical results**
  - Rates reflect structure, intrinsic dimension, and behavior near zero.
  - Minimax optimal up to log factors for the Euclidean compositional class ($\alpha = 0$).

Rates shown for $\alpha = 0$

function class	$\mathcal{C}^{\beta}(\mathcal{M} \times [0,1]^{d_z}, Q)$	$\mathcal{G}_{\text{comp}}$
Exponent of n (log factors omitted)	$-\frac{\beta}{\beta + d_{\mathcal{M}} + d_z}$	$-\min_{i=0,\ldots,q} \frac{\beta_i^*}{\beta_i^* + t_i}$

Minimax optimal up to log factors (Euclidean, $\alpha = 0$)

At n = 10,000, both DNNs have lower mean loss than the kernel baseline in all four compositional settings and three of four manifold settings.

Appendix

Sparsity in DNN

⭐Sparsity in DNN

- Many papers assume neural network sparsity,
  i.e., few nonzero parameters
  - 📌 e.g., Suzuki (2019), Schmidt-Hieber (2020)
- Why? ⇒ excess risks are typically bounded by:

$$\frac{\text{“Complexity”}}{n} + \text{Bias}^2$$

- Complexity is controlled by metric entropy: log covering or bracketing numbers.
- Metric entropy scales with the number of parameters
  → Sparsity leads to a tighter bound
- **Using tighter bias control (Kohler & Langer, 2021), we achieve fast rates without requiring sparsity**